Syllabus:

Data link layer: Error Detection and Correction, Framing, flow and error control, Protocols – Noiseless channels (Simplest, Stop and Wait) and Noisy channels (Stop and Wait and Piggy Backing). Multiple Access Protocols. Random Access-ALOHA, CSMA. Wired LANs-IEEE standards, wireless LANs-Bluetooth, Cellular Telephony

Data link Layer functions include data link control and medium access control.

Data link control functions include *framing, flow and error control, and software implemented protocols* that provide smooth and reliable transmission of frames between nodes.

Error Detection and Correction

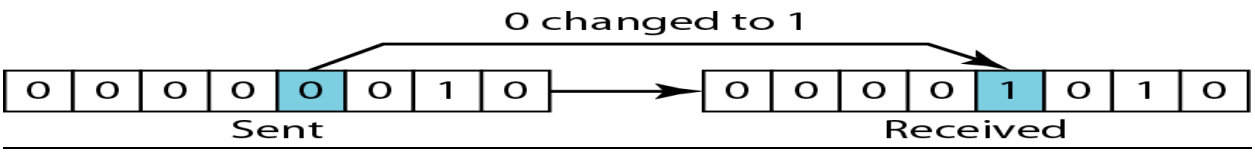
Networks must be able to transfer data from one device to another with acceptable accuracy. For most applications, a system must guarantee that the data received are identical to the data transmitted. Data can be corrupted during transmission. Some applications require that errors be detected and corrected.

Types of Errors

Whenever bits flow from one point to another, they are subject to unpredictable changes because of **interference**. This interference can change the shape of the signal. In a single-bit error, a 0 is changed to a 1 or a 1 to a 0. In a burst error, multiple bits are changed.

Single-Bit Error

The term *single-bit error* means that only 1 bit of a given data unit (such as a byte, character, or packet) is changed from 1 to 0 or from 0 to 1.

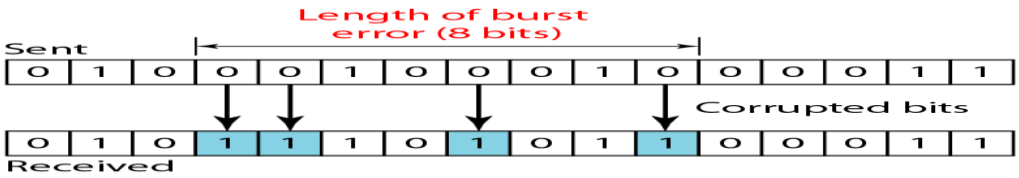


Single-bit errors are the least likely type of error in serial data transmission.

Burst Error

The term *burst error* means that 2 or more bits in the data unit have changed from 1 to 0 or from 0 to 1. Figure shows the effect of a burst error on a data unit. In this case, 0100010001000011 were sent, but 0101110101100011 were received.

Burst error does not mean that the errors occur in consecutive bits. The length of the burst is measured from the first corrupted bit to the last corrupted bit. Some bits in between may not have been corrupted.



Detection versus Correction

Detection

The correction of errors is more difficult than the detection. In error detection, we are looking only to see if any error has occurred. The answer is a simple yes or no. We are not even interested in the number of errors. A single-bit error is the same for us as a burst error.

Correction : In error correction, we need to know the exact number of bits that are corrupted and their location in the message. The number of the errors and the size of the message are important factors.

Forward Error Correction versus Retransmission

There are two main methods of error correction.

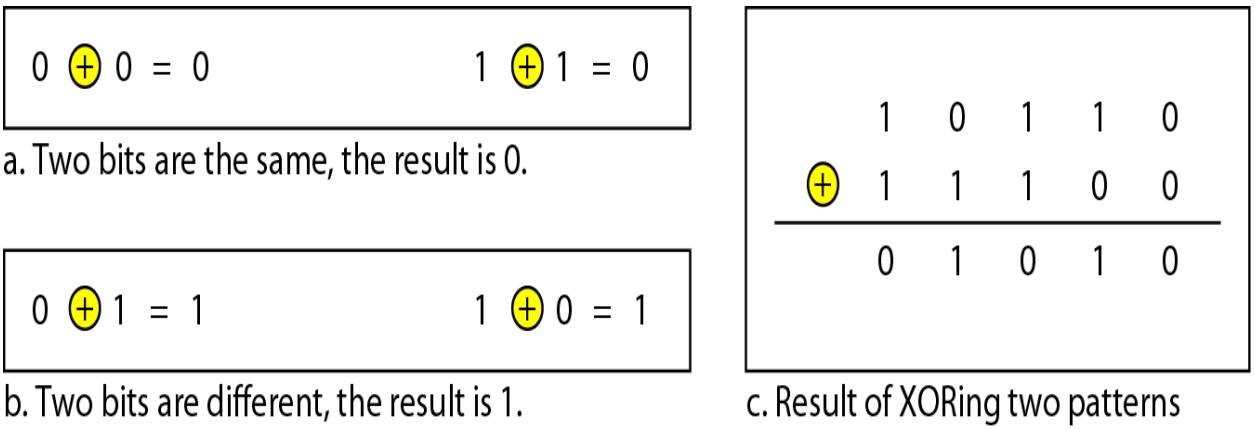
Forward error correction is the process in which the receiver tries to guess the message by using redundant bits. This is possible, if the number of errors is small.

Correction by retransmission: is a technique in which the receiver detects the occurrence of an error and asks the sender to resend the message. Resending is repeated until a message arrives that the receiver believes is error-free (usually, not all errors can be detected).

Coding uses modulo-N arithmetic,

In modulo-N arithmetic, we use only the integers in the range 0 to N −1, inclusive.

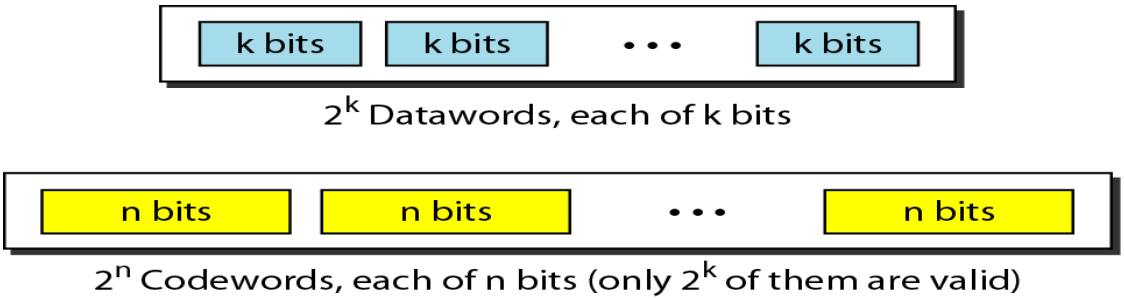
Figure showing XORing of two single bits or two words



BLOCK CODING

In block coding, we divide our message into blocks, each of ***k*** bits, called **datawords**. We add ***r*** redundant bits to each block to make the length ***n = k + r***. The resulting ***n-bit*** blocks are called **codewords.** We have a set of datawords, each of size *k*, and a set of codewords, each of size of *n*. With *k* bits, we can create a combination of *2* datawords; with *n* bits, we can create a combination of *2* codewords. Since *n* > *k*, the number of possible codewords is larger than the number of possible datawords.

The block coding process is one-to-one; the same dataword is always encoded as the same codeword. This means that we have *2* - *2* codewords that are not used.

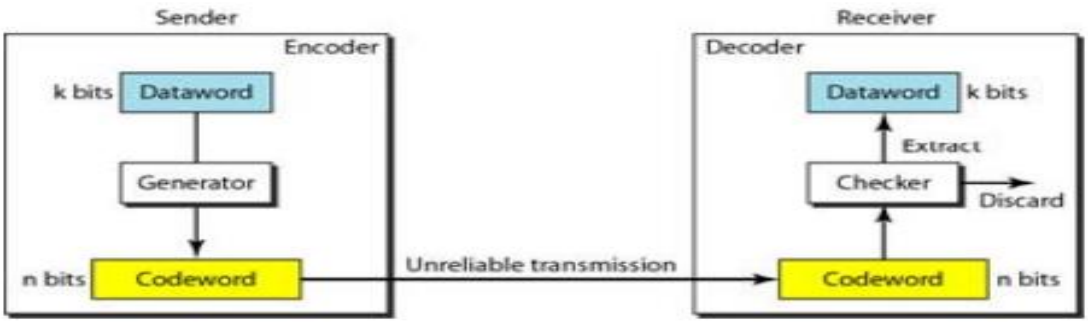


Error Detection

If the following two conditions are met, the receiver can detect a change in the original codeword.

1. The receiver has (or can find) a list of valid codewords.
2. The original codeword has changed to an invalid one.

Figure shows the role of block coding in error detection.



The sender creates codewords out of datawords by using a generator that applies the rules and procedures of encoding. Each codeword sent to the receiver may change during transmission. If the received codeword is the same as one of the valid codewords, the word is accepted; the corresponding dataword is extracted for use. If the received codeword is not valid, it is discarded. However, if the codeword is corrupted during transmission but the received word still matches a valid codeword, the error remains undetected. This type of coding can detect only single errors. Two or more errors may remain undetected.

Example: Let us assume that k =2 and n =3. Table 10.1 shows the list of datawords and codewords.

Table **10.1** *A code for error detection (Example 10.2)*

Datawords	*Codewords*
00	000
01	011
10	101
11	110

Assume the sender encodes the dataword 01 as 011 and sends it to the receiver. Consider the following cases:

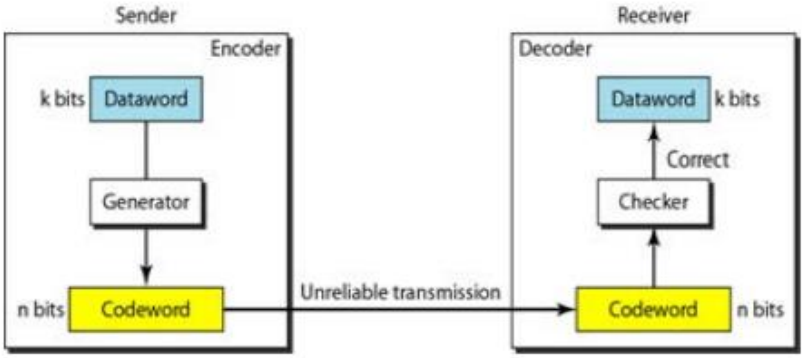
- The receiver receives 011. It is a valid codeword. The receiver extracts the dataword 01 from it.

- The codeword is corrupted during transmission, and 111 is received (the leftmost bit is corrupted).This is not a valid codeword and is discarded.
- The codeword is corrupted during transmission, and 000 is received (the right two bits are corrupted). This is a valid codeword. The receiver incorrectly extracts the dataword 00. Two corrupted bits have made the error undetectable.

An error-detecting code can detect only the types of errors for which it is designed; other types of errors may remain undetected.

Error Correction

Error correction is much more difficult than error detection. In error detection, the receiver needs to know only that the received codeword is invalid; in error correction the receiver needs to find (or guess) the original codeword sent. We can say that we need more redundant bits for error correction than for error detection.



Example:

Assume the dataword is 01. The sender consults the table (or uses an algorithm) to create the codeword 01011. The codeword is corrupted during transmission, and 01001 is received (error in the second bit from the right). First, the receiver finds that the received codeword is not in the table.

This means an error has occurred. (Detection must come before correction.) The receiver, assuming that there is only 1 bit corrupted, uses the following strategy to guess the correct dataword.

LINEAR BLOCK CODE

1. *Simple Parity-Check Code*

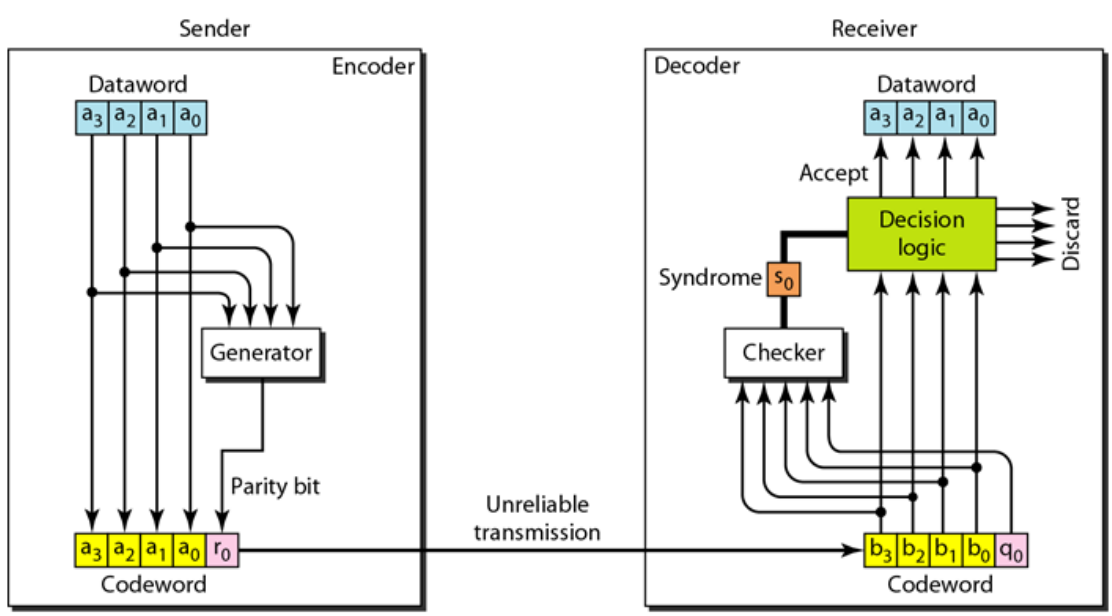
In this code, a *k-bit* dataword is changed to an n-bit codeword where $n = k + 1$. The extra bit, called the **parity bit**, is selected to make the total number of 1s in the codeword even. The minimum Hamming distance for this category is d_{min} =2, which means that the **code is a single-bit error-detecting code; it cannot correct any error.**

Table 10.1 is a parity-check code with k = 2 and n =3. Example: a parity-check code with k= 4 and n =5.

Table 10.3 ***Simple parity-check code C(5, 4)***

Datawords	Codewords	Datawords	Codewords
0000	00000	1000	10001
0001	00011	1001	10010
0010	00101	1010	10100
0011	00110	1011	10111
0100	01001	1100	11000
0101	01010	1101	11011
0110	01100	1110	11101
0111	01111	1111	11110

Encoder and Decoder for parity check



The encoder uses a generator that takes a copy of a 4-bit dataword *(a0,* a1 *a2 a3)* and generates a **parity bit** r_0. The dataword bits and the parity bit create the 5-bit codeword. The parity bit that is added makes the number of 1s in the codeword even.

If the number of 1s is even, parity bit is 0; if the number of 1s is odd, the parity bit is 1. In both cases, the total number of 1s in the codeword is even. It is got by adding all the bits of dataword.

$$r_0 = a_3 + a_2 + a_1 + a_0$$

The sender sends the codeword which may be corrupted during transmission. The receiver receives a 5-bit word. The checker at the receiver will add all 5 bits. The result, which is called the **syndrome,** is just 1 bit.

$$s_0 = b_3 + b_2 + b_1 + b_0 + q_0$$

The syndrome is 0 when the number of 1s in received codeword is even; else it is 1.

Assume the sender sends the dataword 1011. The codeword created from this dataword is 10111, which is sent to the receiver. We examine five cases:

- No error occurs; the received codeword is 10111. The syndrome is 0. The dataword 1011 is created.
- One single-bit error changes a_1.The received codeword is 10011. The syndrome is 1. No dataword is created.
- One single-bit error changes *ro.* The received codeword is 10110. The syndrome is 1. No dataword is created. Here no dataword bits are corrupted, no dataword is created because the code is n`1ot sophisticated enough to show the position of the corrupted bit.
- An error changes *ro* and a second error changesa_3. The received codeword is 00110. The syndrome is 0. The dataword 0011 is created at the receiver. Note that here the dataword is wrongly created due to the syndrome value. The *simple parity-check decoder cannot detect an even number of errors*. The errors cancel each other out and give the syndrome a value of 0.
- Three bits-a_3, a_2, and a_1-are changed by errors. The received codeword is 01011. The syndrome is 1. The dataword is not created.

This shows that the simple parity check, guaranteed to detect one single error, can also find any odd number of errors.

CYCLIC CODES

Cyclic codes are special linear block codes with one extra property. In a cyclic code, if a codeword is cyclically shifted (rotated), the result is another codeword. For example, if 1011000 is a codeword and we cyclically left-shift, then 0110001 is also a codeword.

Framing

Framing in the data link layer separates a message from one source to a destination, or from other messages to other destinations, by adding a sender address and a destination address.

The destination address defines where the packet is to go; the sender address helps the recipient acknowledge the receipt. Frames can be of fixed or variable size.

Fixed-Size Framing

Frames can be of fixed or variable size. In fixed-size framing, there is no need for defining the boundaries of the frames; the size itself can be used as a delimiter.

An example is the ATM wide-area network, which uses frames of fixed size called **cells.**

ATM: Asynchronous Transfer Mode (connection oriented, high-speed network technology that is used in both LAN and WAN over optical fiber and operates upto gigabit speed.

Variable-Size Framing

Variable-size framing is prevalent in local area networks.

In variable-size framing, we need a way to define the end of the frame and the beginning of the next.

Two approaches were used for this purpose:

a) Character-oriented approach:

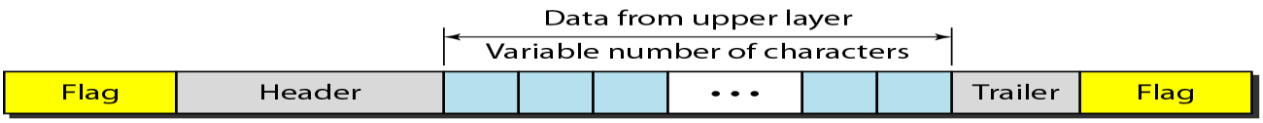


Fig. A frame in character oriented protocol

In a character-oriented protocol, data to be carried are 8-bit characters from a coding system such as ASCII. It also contains

- Header :- which normally carries the source and destination addresses and other control information.
- Trailer :- which carries error detection or error correction redundant bits, are also multiples of 8 bits. To separate one frame from the next.
- Flag :- An 8 bit(1 byte) is added at the beginning and the end of a frame.

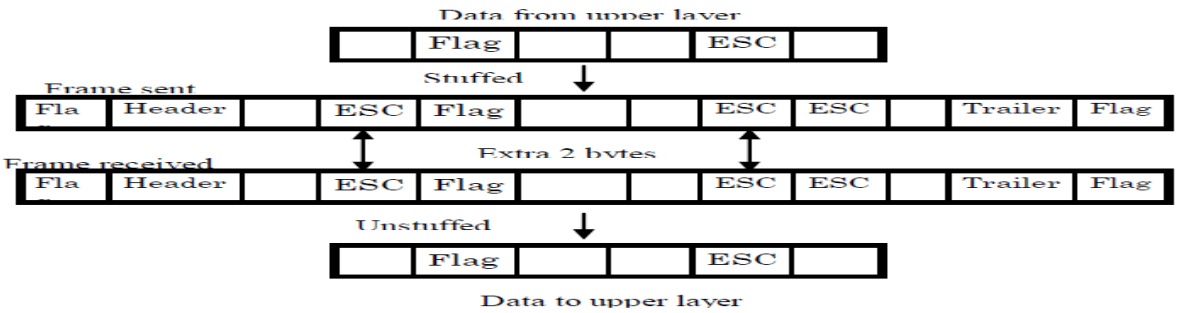


Fig B.*Byte Stuffing & De-stuffing*

b) Bit-oriented approach:

In a bit-oriented protocol, the data section of a frame is a sequence of bits to be interpreted by the upper layer. Most protocols use a special 8-bit pattern flag 01111110 as the delimiter to define the beginning and the end of the frame, as shown in Figure C.

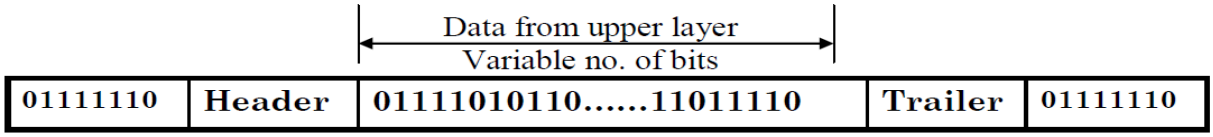


Fig C. *A frame in Bit-Oriented Protocol*

if the flag pattern appears in the data, we have to inform the receiver that this is not the end of the frame. This is done by *stuffing 1 single bit* (instead of 1 byte) to prevent the pattern from looking like a flag. The strategy is called **bit stuffing**.

In bit stuffing, if a 0 and five consecutive I bits are encountered, an extra 0 is added. This extra stuffed bit is eventually removed from the data by the receiver.

Figure D shows bit stuffing at the sender and bit removal at the receiver. Note that even if we have a 0 after five 1s, we still stuff a 0. The 0 will be removed by the receiver.

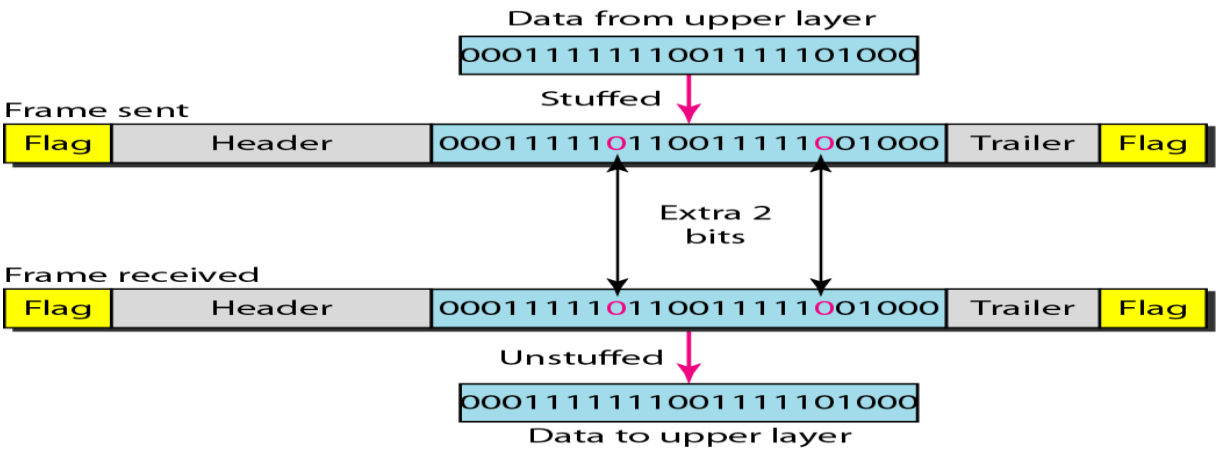


Fig. D. Bit stuffing and unstuffing.

The real flag 01111110 is not stuffed by the sender and is recognized by the receiver.

FLOW AND ERROR CONTROL

The most important responsibilities of the datalink layer are flow control and error control. Collectively, these functions are known as *data link control*.

<u>Flow Control</u>:

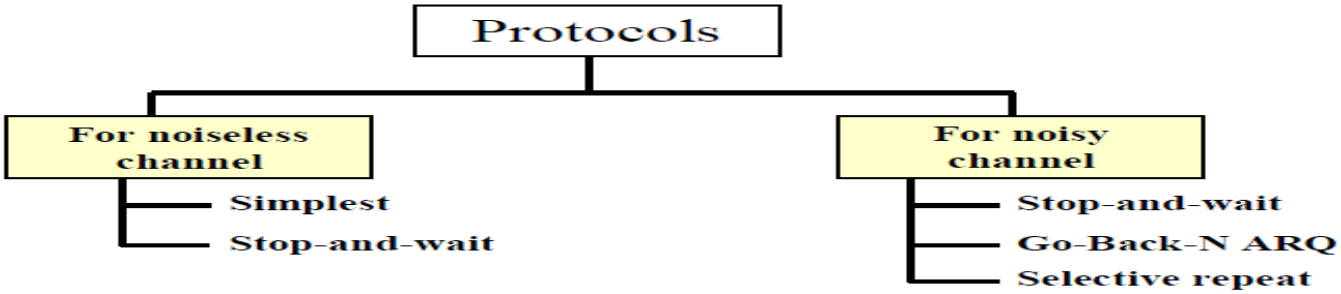
- Any receiving device has a limited speed at which it can process incoming data and a limited amount of memory in which to store incoming data.
- The flow of data must not be allowed to overwhelm the receiver.
- The receiver must be able to tell the sender to halt transmission until it is once again able to receive.

- Flow control refers to a set of procedures used to restrict the amount of data that the sender can send before waiting for acknowledgement.
- Each receiving device has a block of memory, called a **buffer,** reserved for storing incoming data until they are processed. **If** the buffer begins to fill up, the receiver must be able to tell the sender to halt transmission until it is once again able to receive**.**

Error Control:

- Errors occur due to noises in the channel.
- Error control is both error detection & error correction.
- It allows the receiver to tell the sender of any frames lost or damaged in transmission and coordinates the retransmission of those frames by the sender.
- Error control in the data link layer is based on **automatic repeat request**, (ARQ) which is the retransmission of data.

PROTOCOLS: The protocols in the data link layer are classified as:-



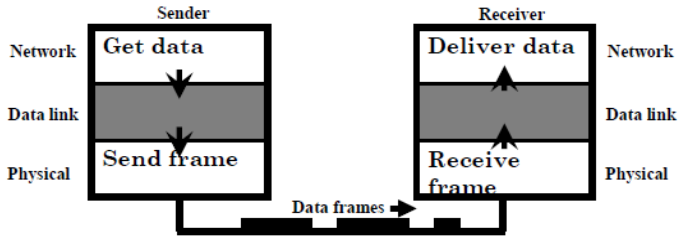
A) Noiseless channels:

i) Simplest protocol:

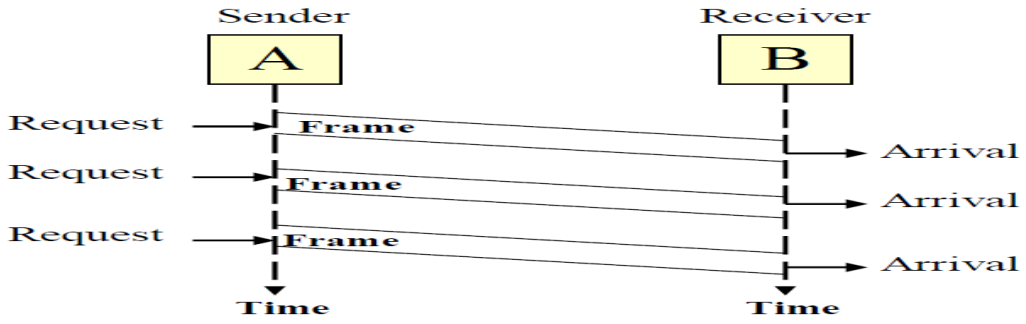
Assumptions in this protocol are:

- Data transfer is unidirectional.
- Both sender & receiver network layers are always ready.
- Processing time can be ignored.
- Infinite buffer space is available.
- Frames are never damaged or lost.

Design :



There is no need for flow control in this scheme. The data link layer at the sender site gets data from its network layer, makes a frame out of the data, and sends it. The datalink layer at the receiver site receives a frame from its physical layer, extracts data from the frame, and delivers the data to its network layer. The data link layers use the services provided by their physical layers (such as signaling, multiplexing, and so on) for the physical transmission of bits. **Fig E. Flow diagram to illustrate simplest protocol**



ii) Stop - & - Wait Protocol:

Assumptions in this protocol are:

- Data transfer is unidirectional.
- Both sender & receiver network layers are always ready.
- Receiver doesn't have enough storage space.
- Receiver is slower than sender in processing.
- Frames are never damaged or lost.

Design :-

We can see the traffic on the forward channel (from sender to receiver) and the reverse channel. At any time, there is either one data frame on the forward channel or one ACK frame on the reverse channel. We therefore need a half-duplex link.

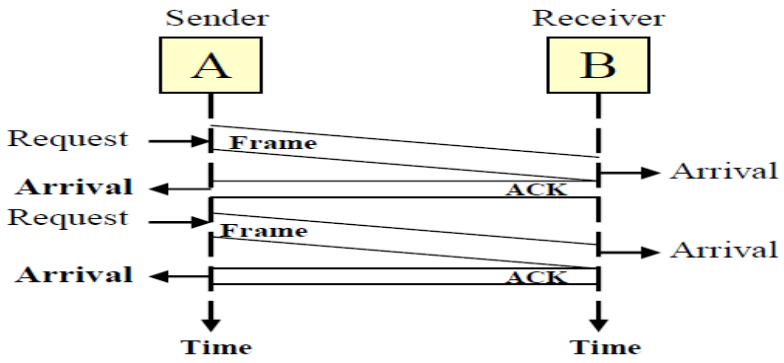


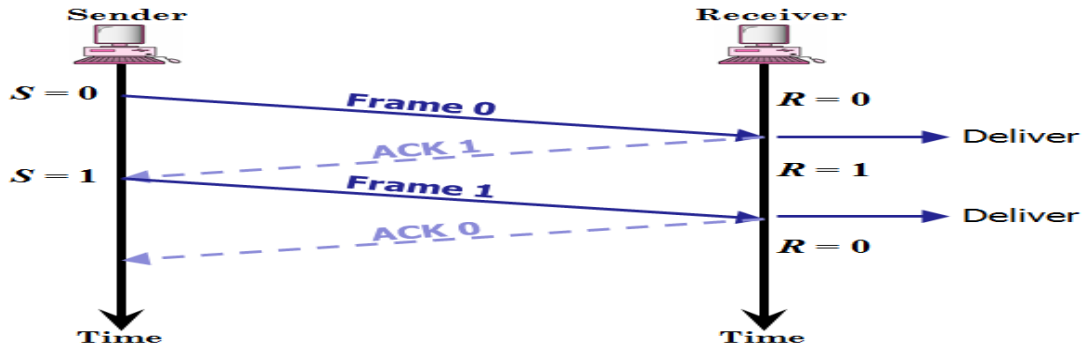
Fig G. Flow diagram to illustrate stop and wait protocol

<u>Noisy channels</u>:-

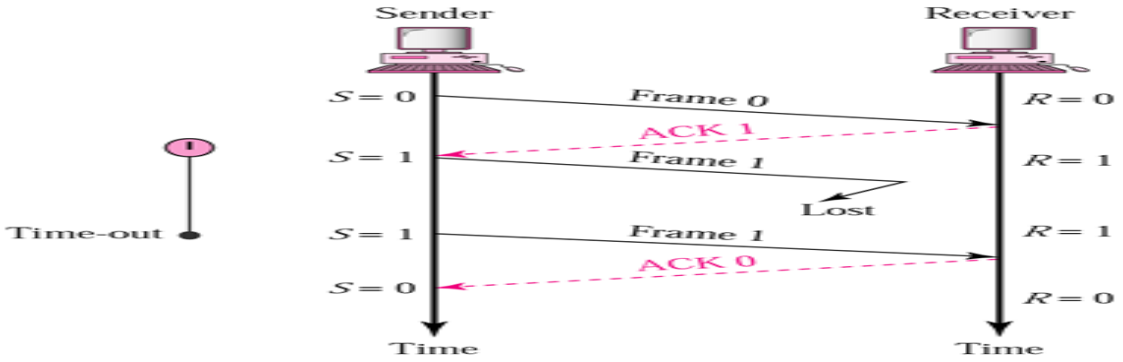
i) Stop - & - Wait Automatic Repeat Request (ARQ):

- Adds a simple error control mechanism to the Stop-&-Wait protocol.
- Error detection & retransmission is used for error control.
- Sending device keeps a copy of the last frame transmitted until acknowledgement for that frame is received.
- Data frames uses Sequence numbers, to avoid duplication of frames.
- For a m bit field range of sequence numbers is from 0 to $2^m - 1$
- ACK frame uses Acknowledgement number, & always announces the sequence number of the next frame expected.
- Timers are used in case of loss of ACK frames.

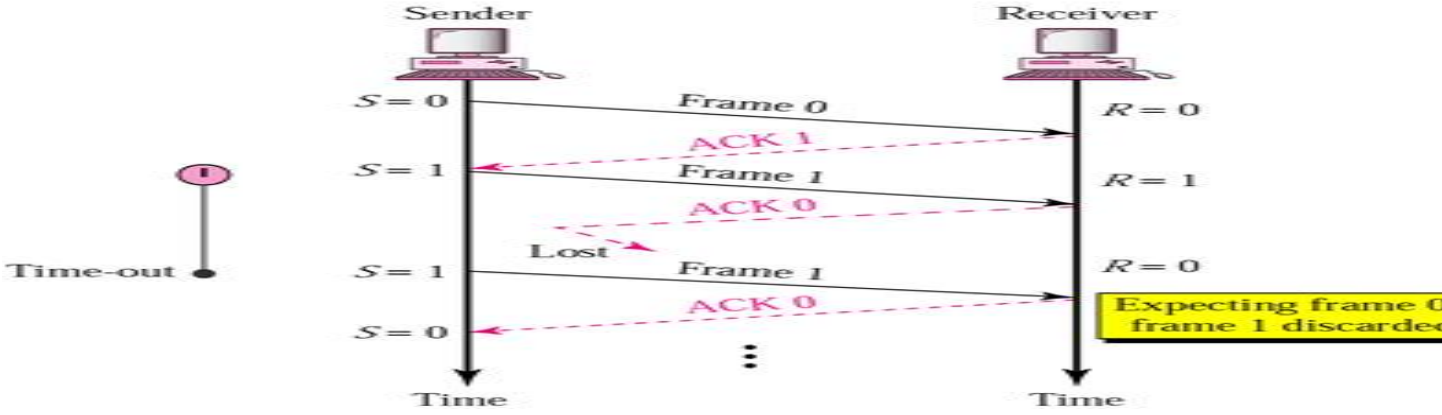
Flow Diagram: Normal operation



Flow Diagram: Lost Frame



Flow Diagram: Lost ACK Frame



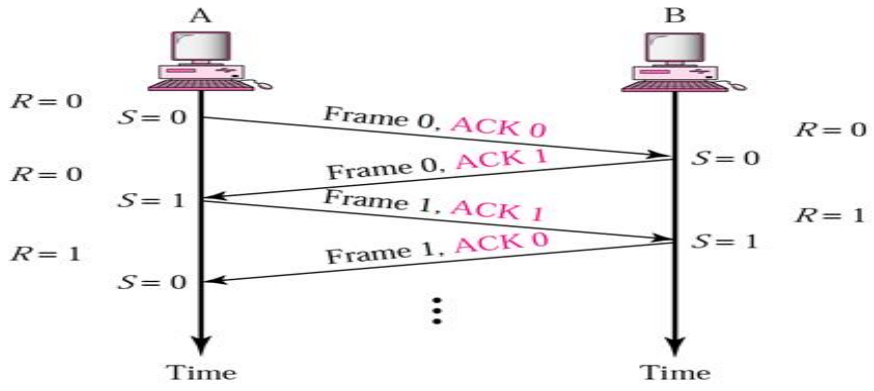
<u>Disadvantages of Stop and Wait ARQ</u>

- In stop-and-wait, at any point in time, there is only one frame that is sent and waiting to be acknowledged.
- This is not a good use of transmission medium.
- To improve efficiency, multiple frames should be in transition while waiting for ACK.
- Two protocol use the above concept,
  - **Go-Back-N ARQ**
  - **Selective Repeat ARQ**

PIGGYBACKING

The three protocols Stop and Wait ,Go-Back-N ARQ Selective Repeat ARQ are all unidirectional: data frames flow in only one direction although control information such as ACK and NAK frames can travel in the other direction. In real life, data frames are normally flowing in both directions: from node A to node B and from node B to node A. This means that the control information also needs to flow in both directions.

A technique called **piggybacking** is used to improve the efficiency of the bidirectional protocols. When a frame is carrying data from A to B, it can also carry control information about arrived (or lost) frames from B; when a frame is carrying data from B to A, it can also carry control information about the arrived (or lost) frames from A.

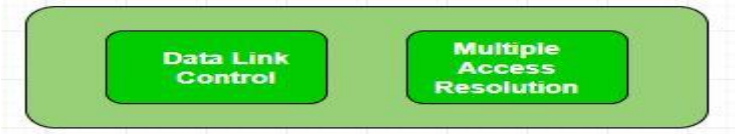


Multiple Access Protocols

The Data Link Layer is responsible for transmission of data between two nodes. Its main functions are-

- Data Link Control

- Multiple Access Control

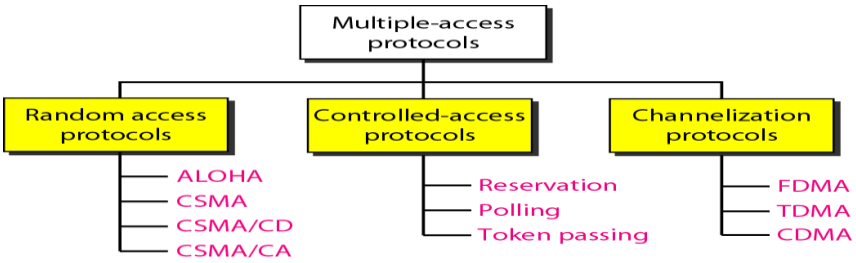


Data Link control

The data link control is responsible for reliable transmission of message over transmission channel by using techniques like *framing, error control and flow control*.

Multiple Access Control

If there is a dedicated link between the sender and the receiver then data link control layer is sufficient, however if there is no dedicated link present then multiple stations can access the channel simultaneously. Hence multiple access protocols are required to decrease collision and avoid crosstalk. For example, in a classroom full of students, when a teacher asks a question and all the students (or stations) start answering simultaneously (send data at same time) then a lot of chaos is created (data overlap or
data lost) then it is the job of the teacher (multiple access protocols) to manage the students and make them answer one at a time.
Thus, protocols are required for sharing data on non-dedicated channels. Multiple access protocols can be subdivided further as –



1. Random Access Protocol: In this, all stations have same superiority that is no station has more priority than another station. Any station can send data depending on medium's state (idle or busy). It has two features:

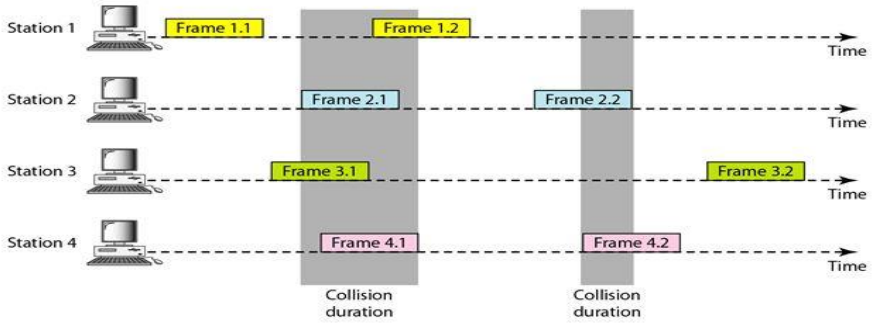
1. There is no fixed time for sending data
2. There is no fixed sequence of stations sending data

The Random access protocols are further subdivided as:

(a) ALOHA – It was designed for wireless LAN but is also applicable for shared medium. In this, multiple stations can transmit data at the same time and can hence lead to collision and data being garbled.

Pure Aloha:

The original ALOHA protocol is called pure ALOHA. Each station sends a frame whenever it has a frame to send. Since there is only one channel to share, there is the possibility of collision between frames from different stations. The following figure shows an example of frame collisions in pure ALOHA.



Only two frames survive: frame 1.1 from station 1 and frame 3.2 from station 3. We need to mention that even if one bit of a frame coexists on the channel with one bit from another frame, there is a collision and both will be destroyed.

When a station sends data it waits for an acknowledgement. If the acknowledgment does not arrive after a time-out period, the station assumes that the frame (or the acknowledgment) has been destroyed and resends the frame.

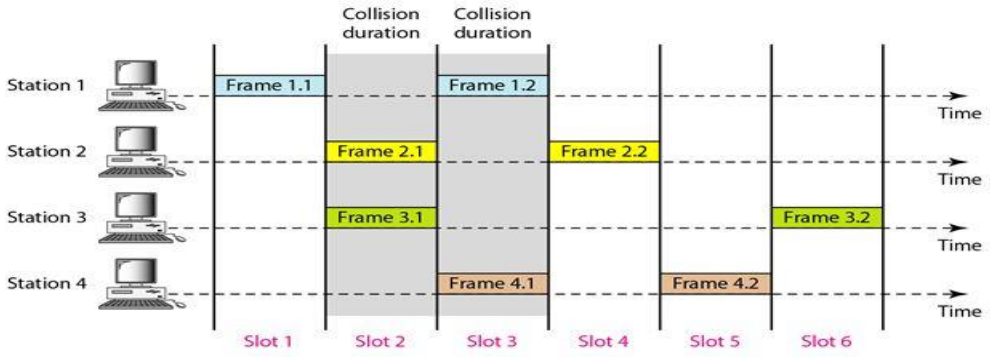
A collision involves two or more stations. If all these stations try to resend their frames after the time-out, the frames will collide again.

Pure ALOHA says that when the time-out period passes, each station waits a random amount of time before resending its frame. The randomness will help to avoid more collisions.

Slotted ALOHA

Pure ALOHA has a vulnerable time of 2 x T_{FR}. Because there is no rule that defines when the station can send. A station may send soon after another station has started or soon before another station has finished. Slotted ALOHA was invented to improve the efficiency of pure ALOHA.

In slotted ALOHA we divide the time into slots of T_{fr} s and force the station to send only at the beginning of the time slot. The following figure shows an example of frame collisions in slotted ALOHA.



Because a station is allowed to send only at the beginning of the synchronized time slot, if a station misses this until the beginning of the next time slot. This means that the station which started at the beginning of this slot h sending its frame. But, there is the possibility of collision if two stations try to send at the beginning of the same

Carrier Sense Multiple Access (CSMA)

To minimize the chance of collision and to, increase performance the CSMA method was developed. The chance of collision can be reduced if a station senses a medium before trying to use it.

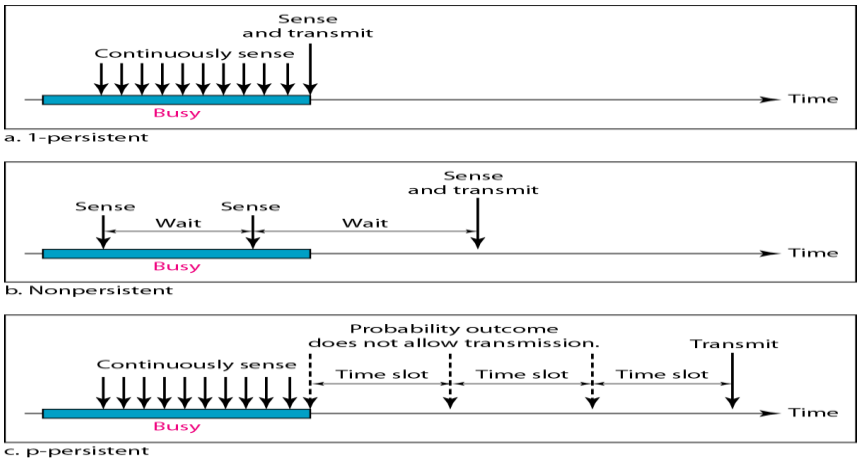
Carrier sense multiple access (CSMA) requires that each station first listen to the medium (or check the state of the medium before sending. CSMA is based on the principle "sense before transmit" or "listen before talk." It cannot eliminate collision.

PERSISTENCE METHODS

What should a station do if the channel is busy? What should a station do if the channel is idle? Three methods have been used:

- the 1-persistent method,
- the non persistent method, and
- the p-persistent method.

Figure 12.10 shows the behavior of three persistence methods when a station finds a channel busy.



1-Persistent: The **1-persistent method** is simple and straightforward. In this method after the station finds the line idle, it sends its frame immediately (with probability 1). This method has the highest chance of collision because two or more stations may find the line idle and send their frames immediately.

Non persistent: In the **Non persistent method,** a station that has a frame to send senses the line. If the line is idle, it sends immediately. If the line is not idle it waits random amount of time and then senses the line again, the non persistent approach reduces the chance of collision because there is less chance that two or more stations will wait the same amount of time and retry to send simultaneously. But, this method reduces the efficiency of the network because the medium remains idle when there may be stations with frames to send.

P-Persistent method is used if the channel has time slots with slot duration equal to or greater than the maximum propagation time. The p-persistent approach combines the advantages of the other two methods; it reduces the chance of collision and improves efficiency. In this method after the station finds the line idle it follows these steps:

1. With probability p, the station sends its frame.

2. With probability *q =1* - p. the station waits for the beginning of the next time slot and checks the line again.

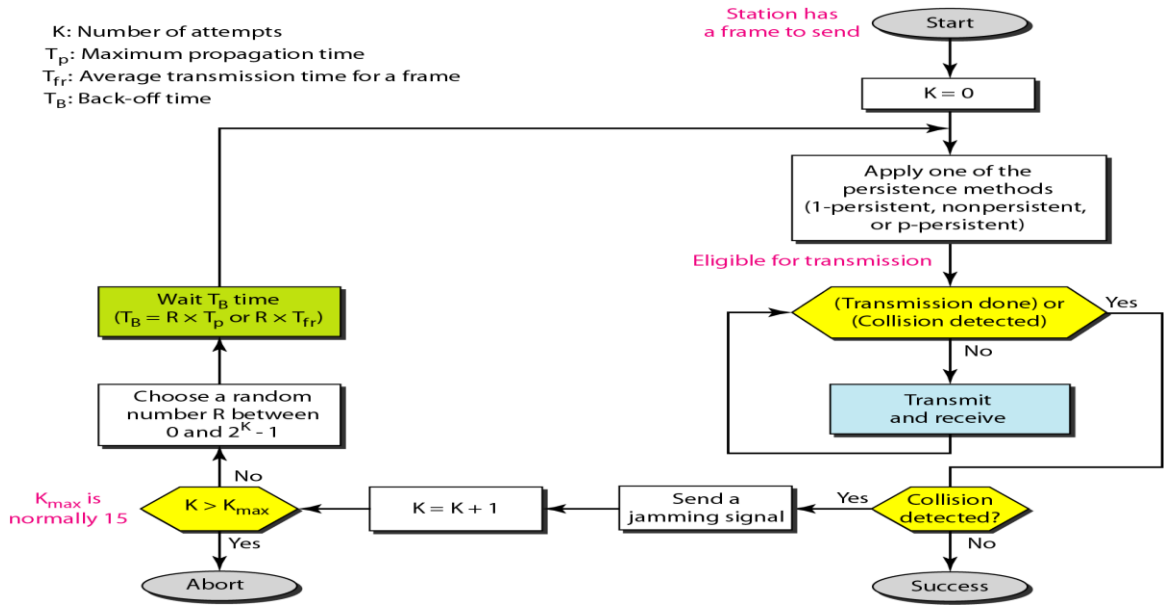
a. If the line is idle. It goes to step 1.

b. If the line is busy. It acts as though a collision has occurred and uses the back off procedure.

CSMA/CD (Carrier Sense Multiple Access/Collision Detection)

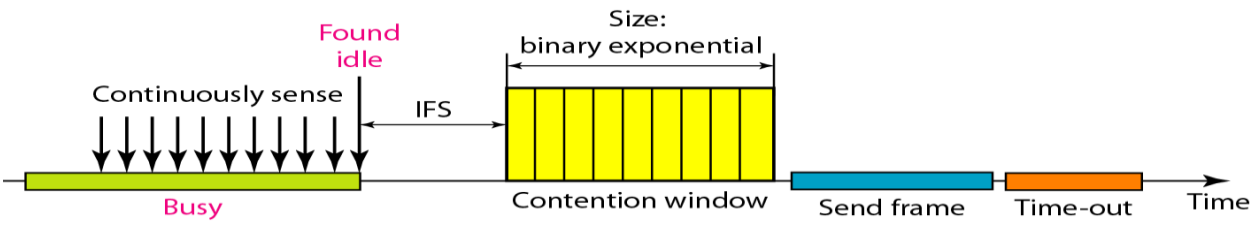
If a collision is detected during transmission of a packet, the node immediately stops transmission and it transmits *jamming signal* for a brief duration to ensure that all stations know that collision has occurred.

After transmitting the jamming signal, the node waits for a random amount of time and then transmission is resumed. The random delay ensures that the nodes, which were involved in the collision, are not likely to have a collision at the time of retransmissions. To achieve stability in the back off scheme, a technique known as *binary exponential back off* is used. A node will attempt to transmit repeatedly in the face of repeated collisions, but after each collision, the mean value of the random delay is doubled. After 15 retries (excluding the original try), the unlucky packet is discarded and the node reports an error. A flowchart representing the binary exponential back off algorithm is given in Fig. 5.2.11.



CSMA/CA(*Carrier Sense Multiple Access/Collision Avoidance)*

CSMA/CA protocol is used in wireless networks because they cannot detect the collision so the only solution is collision avoidance.



• CSMA/CA avoids the collisions using three basic techniques.

(i) Interframe space

(ii) Contention window

(iii) Acknowledgements

1. Interframe Space (IFS)

• Whenever the channel is found idle, the station does not transmit immediately. It waits for a period of time called **interframe space (IFS).**

• When channel is sensed to be idle, it may be possible that same distant station may have already started transmitting and the signal of that distant station has not yet reached other stations.

• Therefore the purpose of IFS time is to allow this transmitted signal to reach other stations.

• If after this IFS time, the channel is still idle, the station can send, but it still needs to wait a time equal to contention time.

• IFS variable can also be used to define the priority of a station or a frame.

2. Contention Window

• Contention window is an amount of time divided into slots.

• A station that is ready to send chooses a random number of slots as its wait time.

• The number of slots in the window changes according to the binary exponential back-off strategy. It means that it is set of one slot the first time and then doubles each time the station cannot detect an idle channel after the IFS time.

• This is very similar to the p-persistent method except that a random outcome defines the number of slots taken by the waiting station.

• In contention window the station needs to sense the channel after each time slot.

• If the station finds the channel busy, it does not restart the process. It just stops the timer & restarts it when the channel is sensed as idle.

Wireless LAN ----Bluetooth

It is a wireless LAN technology designed to connect devices of different functions such as telephones, notebooks, computers, cameras, printers, coffee makers, and so on. A Bluetooth LAN is an ad hoc network, which means that the network is formed spontaneously.

Bluetooth Architecture

Bluetooth architecture defines two types of networks:

1. Piconet

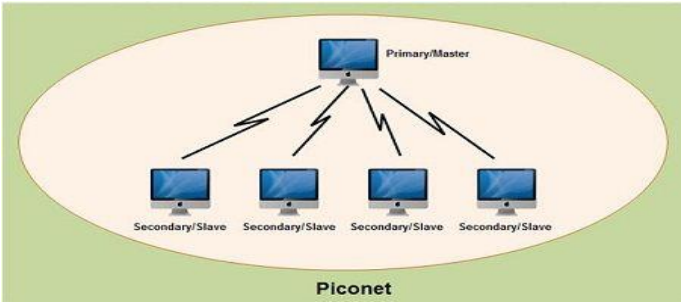
2. Scattemet

1. Piconet

• Piconet is a Bluetooth network that consists of one primary (master) node and seven active secondary (slave) nodes.

• Thus, piconet can have upto eight active nodes (1 master and 7 slaves) or stations within the distance of 10 meters.

• The communication between the primary and the secondary can be one-to-one or one-to-many.



• All communication is between master and a slave. Salve-slave communication is not possible.

• In addition to seven active slave station, a piconet can have up to 255 parked nodes. These parked nodes are secondary or slave stations and cannot take part in communication until it is moved from parked state to active state.

2. Scatternet

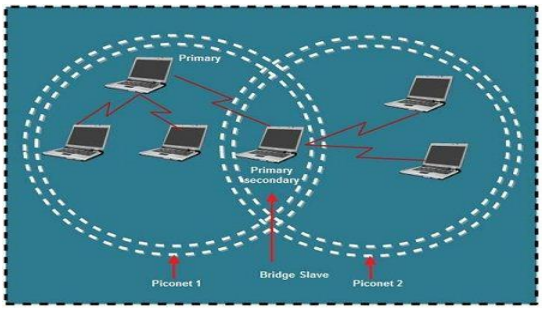
• Scatternet is formed by combining various piconets.

• A slave in one piconet can act as a master or primary in other piconet.

• Such a station or node can receive messages from the master in the first piconet and deliver the message to its slaves in other piconet where it is acting as master. This node is also called *bridge slave*.

• Thus a station can be a member of two piconets.

• A station cannot be a master in two piconets.



Bluetooth devices

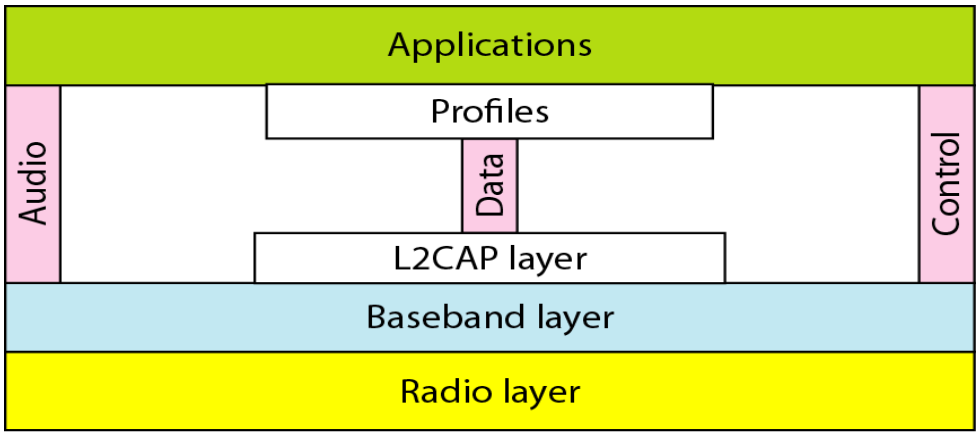
Bluetooth device has a built in short range radio transmitter. The current data rate is 1 Mbps with 2.4 GHZ bandwidth.

Bluetooth layers and Protocol Stack

• Bluetooth standard has many protocols that are organized into different layers.

• The layer structure of Bluetooth does not follow OS1 model, TCP/IP model or any other known model.

• The different layers and Bluetooth protocol architecture.



Radio Layer

• The Bluetooth radio layer corresponds to the physical layer of OSI model.

• It deals with radio transmission and modulation.

• The radio layer moves data from master to slave or vice versa.

• It is a low power system that uses 2.4 GHz ISM band in a range of 10 meters.

•Bluetooth uses the Frequency Hopping Spread Spectrum (FHSS) method in the physical layer to avoid interference from other devices or networks.

• Bluetooth hops 1600 times per second, *i.e.* each device changes its modulation frequency 1600 times per second.

Baseband Layer

• Baseband layer is equivalent to the MAC sublayer in LANs.

• Bluetooth uses a form of TDMA called TDD-TDMA (time division duplex TDMA).

• Master and slave stations communicate with each other using time slots.

• The master in each piconet defines the time slot of 625 µsec.

• In TDD- TDMA, communication is half duplex in which receiver can send and receive data but not at the same time.

• If the piconet has only no slave; the master uses even numbered slots (0, 2, 4 ...) and the slave uses odd-numbered slots (1, 3, 5, ...). Both master and slave communicate in half duplex mode.

In slot 0, master sends & secondary receives; in slot 1, secondary sends and primary receives.(**single secondary communication**)

• If piconet has more than one slave, the master uses even numbered slots. The slave sends in the next odd-numbered slot if the packet in the previous slot was addressed to it.

• In Baseband layer, two types of links can be created between a master and slave. These are:

1. Asynchronous Connection-less (ACL)

• It is used for packet switched data that is available at irregular intervals.

• ACL delivers traffic on a best effort basis. Frames can be lost & may have to be retransmitted.

• A slave can have only one ACL link to its master.

• Thus ACL link is used where correct delivery is preferred over fast delivery.

• The ACL can achieve a maximum data rate of 721 kbps by using one, three or more slots.

2. Synchronous Connection Oriented (SCO)

• sco is used for real time data such as sound. It is used where fast delivery is preferred over accurate delivery.

• In a sco link, a physical link is created between the master and slave by reserving specific slots at regular intervals.

• Damaged packet; are not retransmitted over sco links.

• A slave can have three sco links with the master and can send data at 64 Kbps.

Logical Link, Control Adaptation Protocol Layer (L2CAP)

• The logical unit link control adaptation protocol is equivalent to logical link control sublayer of LAN.

• The ACL link uses L2CAP for data exchange but sco channel does not use it.

• The various function of L2CAP is:

1. Segmentation and reassembly

• L2CAP receives the packets of upto 64 KB from upper layers and divides them into frames for transmission.

• It adds extra information to define the location of frame in the original packet.

• The L2CAP reassembles the frame into packets again at the destination.

2. Multiplexing

• L2CAP performs multiplexing at sender side and demultiplexing at receiver side.

• At the sender site, it accepts data from one of the upper layer protocols frames them and deliver them to the Baseband layer.

• At the receiver site, it accepts a frame from the baseband layer, extracts the data, and delivers them to the appropriate protocol1ayer.

3. Quality of Service (QOS)

• L2CAP(**Logical Link, Control Adaptation Protocol Layer (L2CAP))**

handles quality of service requirements, both when links are established and during normal operation.

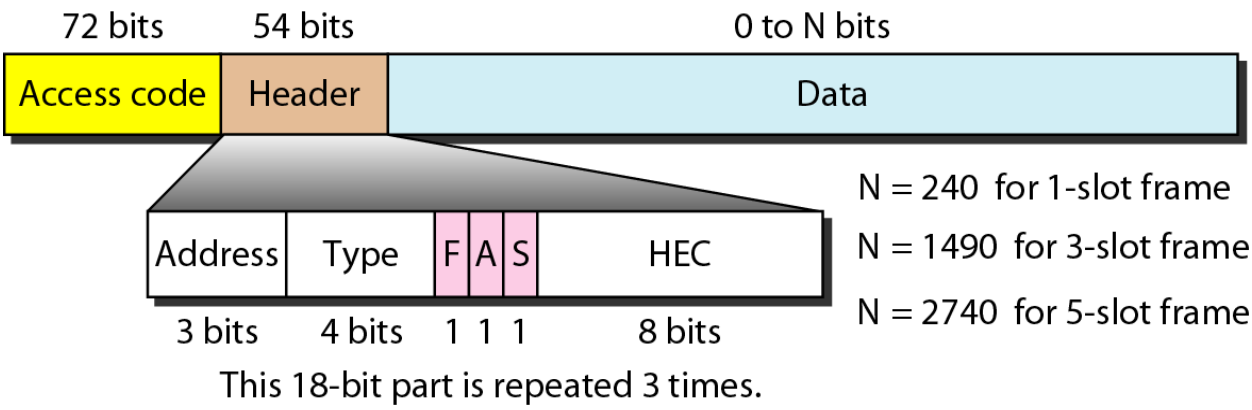
• It also enables the devices to negotiate the maximum payload size during connection establishment.

4. Group Management

It allows devices to create a type of logical addressing between themselves. I.e. To or three secondary devices can be part of a multicast group to receive data from the primary.

Bluetooth Frame Format

The various fields of blue tooth frame format are:



1. **Access Code**: It is 72 bit field that contains synchronization bits. It identifies the master.

2. **Header**: This is 54-bit field which contain 18 bit pattern.

The header field contains following subfields:

(i) **Address**: This 3 bit field can define upto seven slaves (1 to 7). If the address is zero, it is used for broadcast communication from primary to all secondaries.

(ii)Type: This 4 bit field identifies the type of data coming from upper layers.

(iii) **F**: This flow bit is used for flow control. When set to 1, it means the device is unable to receive more frames.

(iv) **A**: This bit is used for acknowledgement. Bluetooth uses stop and wait ARQ.

(v) **S**: This bit contains a sequence number of the frame to detect retransmission.

(vi) **HEC** : This 8 bit field contains checksum to detect errors in 18 bit header section.

3. Payload: This field can be 0 to 2744 bits long. It contains data or control information coming from upper layers.

CELLULAR TELEPHONY

Cellular telephony is a system-level concept, which replaces a single high power transmitter with a large number of low-power transmitters for communication between any two devices over a large geographic area.

Primary goal of the cellular telephone network is to provide wireless communication between

two moving devices, called *mobile stations* or between one mobile unit and a stationary unit, commonly referred to as *land-line* unit.

To accommodate a large number of users over a large geographic area, the cellular telephone system uses a large number of low-power wireless transmitters to create ***cells.*** Variable power levels allow cells to be sized according to subscriber density and demand within a particular region. As mobile users travel from cell to cell, their conversations are handed off between *cells*. //hard handoff; softhandoff

Channels (frequencies) used in one cell can be reused in another cell some distance away, which allows communication by a large number stations using a limited number of radio frequencies. The basic concept of reuse allows a fixed number of channels to serve an arbitrarily large number of users.

Cellular Telephone System

A cellular system comprises the following basic components:

- **Mobile Stations (MS):** Mobile handsets, which is used by an user to communicate with another user
- **Cell**: Each cellular service area is divided into small regions called cell (5 to 20 Km)
- **Base Stations (BS):** Each cell contains an antenna, which is controlled by a small office.
- **Mobile Switching Center (MSC**): Each base station is controlled by a switching office, called mobile switching center

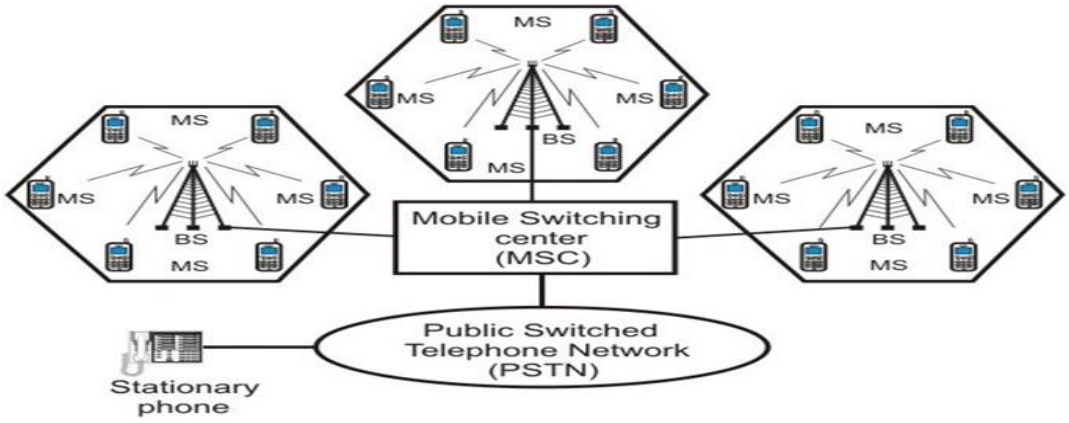


Figure 5.9.1 Schematic diagram of a cellular telephone system

Frequency Reuse Principle

Frequency reusing is the concept of using the same radio frequencies within a given area, that are separated by considerable distance, with minimal interference, to establish communication.

Frequency reuse offers the following benefits −

- Allows communications within cell on a given frequency
- Limits escaping power to adjacent cells
- Allows re-use of frequencies in nearby cells
- Uses same frequency for multiple conversations
- 10 to 50 frequencies per cell

For example, when **N** cells are using the same number of frequencies and **K** be the total number of frequencies used in systems. Then each **cell frequency** is calculated by using the formulae **K/N**.

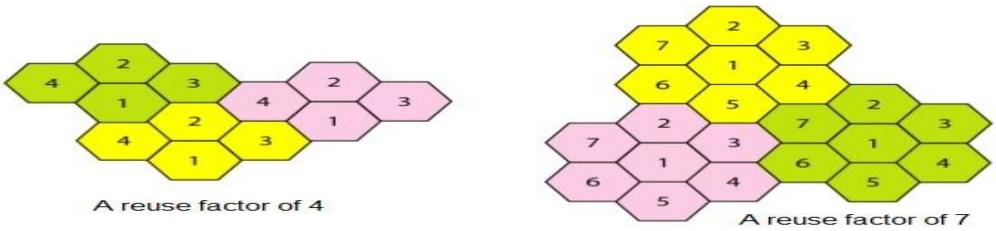


Figure 5.9.2 (a) Cells showing reuse factor of 4 (b) Cells showing reuse factor of 7

<u>Transmitting and Receiving</u>

Transmitting involves the following steps:

- A caller enters a 10-digit code (phone number) and presses the send button.
- The MS scans the band to select a free channel and sends a strong signal to send the number entered.
- The BS relays the number to the MSC.
- The MSC in turn dispatches the request to all the base stations in the cellular system.
- The Mobile Identification Number (MIN) is then broadcast over all the forward control channels throughout the cellular system. It is known as *paging*.
- The MS responds by identifying itself over the reverse control channel.

- The BS relays the acknowledgement sent by the mobile and informs the MSC about the handshake.
- The MSC assigns an unused voice channel to the call & call is established.

Receiving involves the following steps:

- All the idle mobile stations continuously listens to the paging signal to detect messages directed at them.
- When a call is placed to a mobile station, a packet is sent to the callee's home MSC to find out where it is.
- A packet is sent to the base station in its current cell, which then sends a broadcast on the paging channel.
- The callee MS responds on the control channel.
- In response, a voice channel is assigned and ringing starts at the MS.

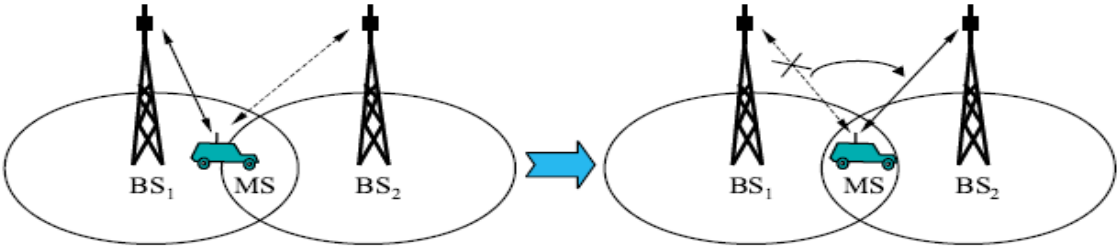
Mobility Management

A MS is assigned a home network, commonly known as location area. When an MS migrates out of its current BS into the footprint of another, a procedure is performed to maintain service continuity, known as *Handoff management*. An agent in the home network, called *home agent*, keeps track of the current location of the MS. The procedure to keep track of the user's current location is referred to as *Location management*. Handoff management and location management together are referred to as *Mobility management*.

Handoff: At any instant, each mobile station is logically in a cell and under the control of the cell's base station. When a mobile station moves out of a cell, the base station notices the MS's signal fading away and requests all the neighbouring BSs to report the strength they are receiving. The BS then transfers ownership to the cell getting the strongest signal and the MSC changes the channel carrying the call. The process is called ***handoff.***

There are two types of handoff; Hard Handoff and Soft Handoff.

In a ***hard handoff***, which was used in the early systems, a MS communicates with one BS. As a MS moves from cell A to cell B, the communication between the MS and base station of cell A is first broken before communication is started between the MS and the base station of B. As a consequence, the transition is not smooth.



For smooth transition from one cell (say A) to another (say B), an MS continues to talk to both A and B. As the MS moves from cell A to cell B, at some point the communication is broken with the old base station of cell A. This is known as ***soft handoff*.**

Roaming: Two fundamental operations are associated with Location Management; *location update* and *paging*. When a Mobile Station (MS) enters a new Location Area, it performs a location updating procedure by making an association between the foreign agent and the home agent. One of the BSs, in the newly visited Location Area is informed and the home directory of the MS is updated with its current location. When the home agent receives a message destined for the MS, it forwards the message to the MS via the foreign agent. An authentication process is performed before forwarding the message.

First Generation System

It was designed for voice communication. One example is Advanced Mobile Phone System (AMPS) used in North America. An AMP is an analog cellular phone system.

It uses 800 MHz ISM band and two separate analog channels; forward and reverse analog channels. The band between 824 to 849 MHz is used for reverse communication from MS to BS. The band between 869 to 894 MHz is used for forward communication from BS to MS. Each band is divided in to 832 30-KHz channels as shown in Fig. 5.9.8. As each location area is shared by two service providers, each provider can have 416 channels, out of which 21 are used for control. AMPS uses Frequency Division Multiple Access (FDMA) to divide each 25-MHz band into 30-KHz channels as shown in Fig. 5.9.9.

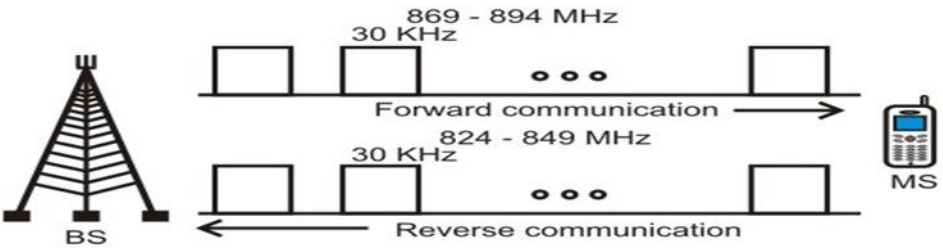
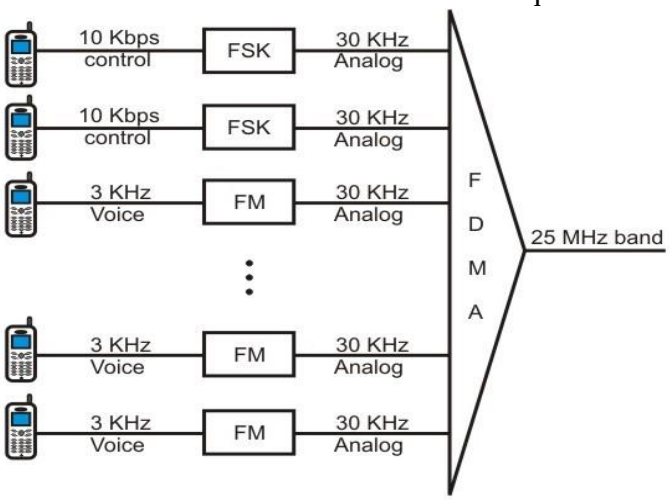


Figure 5.9.9 FDMA medium access control technique used in AMPS



Second Generation

The first generation cellular network was developed for analog voice communication. To provide better voice quality, the second generation was developed for digitized voice communication. Three major systems were evolved, as shown in Fig. 5.9.10.

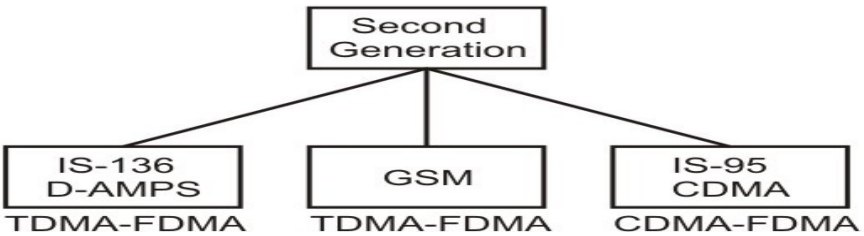


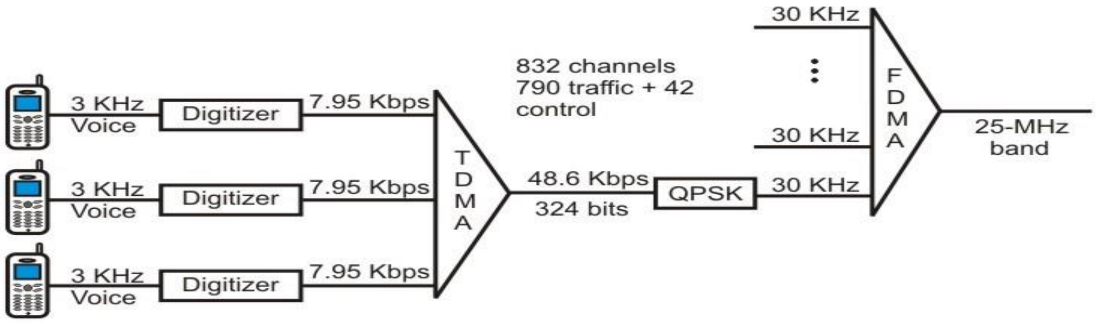
Figure 5.9.10 FDMA medium access control technique used in AMPS

D-AMPS: D-AMPS is essentially a digital version of AMPS and it is backward compatible with AMPS. It uses the same bands and channels and uses the frequency reuse factor of 1/7. 25 frames per second each of 1994 bits, divided in 6 slots shared by three channels. Each slot has 324 bits-159 data, 64 control, 101 error-correction bits as shown in Fig. 5.9.11. As shown in the

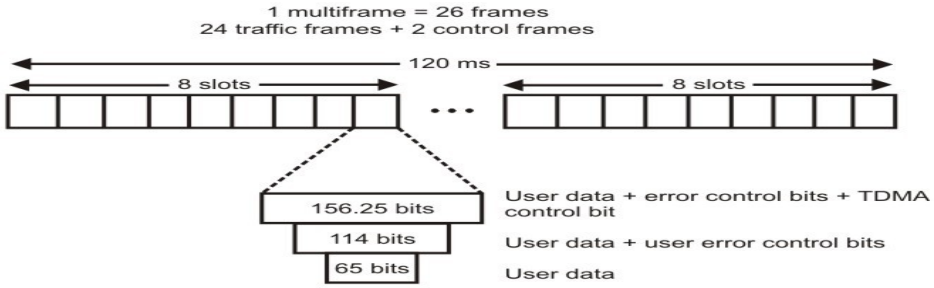
figure, it uses both TDMA and FDMA medium access control techniques.

Figure 5.9.11 D-MPS

GSM: The Global System for Mobile (GSM) communication is a European standard developed to replace the first generation technology. It uses two bands for duplex communication. Each voice channel is digitized and compressed to a 13Kbps digital signal. Each slot carries 156.25



bits, 8 slots are multiplexed together creating a FDM frame, 26 frames are combined to form a



multiframe, as shown in Fig. 5.9.12. For medium access control, GSM combines both TDMA and FDMA. There is large amount of overhead in TDMA, 114 bits are generated by adding extra bits for error correction. Because of complex error correction, it allows a reuse factor as low as 1/3.

Figure 5.9.12 Multiframe components

IS-95: IS-95 is based on CDMA/DSSS and FDMA medium access control technique. It uses the forward and backward transmissions.

Third Generation

Goals of the third generation (3G) technologies are mentioned below:

- Allow both digital data and voice communication.

- To facilitate universal personnel communication.
- Listen music, watch movie, access internet, video conference, etc. Criteria for 3G Technologies are:
- Voice quality: Same as present PSTN network.
- Data rate: 144Kbps (car), 384 (pedestrians) and 2Mbps (stationary).
- Support for packet-switched and circuit-switched data services.
- Bandwidth of 2 MHz.
- Interface to the internet.

ITU developed a blueprint called Internet Mobile Communication for year 2000 (IMT-2000). All five Radio Interfaces adopted by IMT-2000 evolved from the second generation technologies as shown in Fig. 5.9.15.

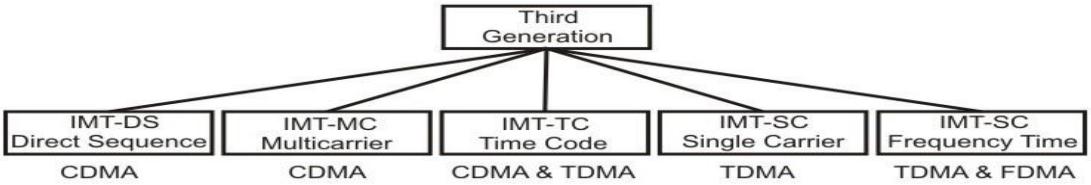


Figure 5.9.15 Third generation cellular technologies